



Golam Mohiuddin Shaifullah

Contact

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Personal Details

Born: December 21, 1985—Kolkata, India
Nationality: Indian

Current position

Post-doctoral Researcher, Dipartimento di Fisica 'G. Occhialini', Università degli Studi di Milano-Bicocca.
Supervisor: Prof. Alberto SESANA
Start date: 10/10/2020

I build **high-precision datasets** for **pulsar timing arrays (PTAs)**, which detect **low-frequency gravitational waves** from merging **supermassive black holes**. My work combines data from multiple telescopes within the **European** and **International PTAs**, investigates **interstellar medium** effects on pulsar signals, and refines **modeling techniques** to boost detection prospects.

I explore **pulsar properties** across a broad radio frequency range, using major observatories such as **Arecibo**, **Effelsberg**, **GMRT**, **Parkes**, **LOFAR**, **NenuFAR**, and **MeerKAT**.

I am an active member of the **European** and **International PTA** collaborations. Beyond research, I serve as **Secretary General** of the Gravitational Wave Early Career Scientists group, founded the **International PTA Early Career Scientists** working group, and remain committed to **science education** and an **inclusive** academic community.

Appointments held

- **2017–2020**: Postdoctoral Researcher, ASTRON, the Netherlands.
- **2012–2013**: Senior Research Fellow, University of Pune, India.
- **2010–2012**: Junior Research Fellow, University of Pune, India.

Education

- **2013–2017**: Dr. rer. Nat. (PhD), Universität Bielefeld, Bielefeld, Germany. **Date of degree: 10/05/2017**
- **2008–2010**: MSc (Physics), FIRST CLASS, Fergusson College, University of Pune, Pune, India
- **2005–2008**: BSc (Physics, Electronics, Mathematics), FIRST CLASS, St Joseph's College, Bangalore University, Bangalore, India

Thesis

- **May, 2017**: **Timing and properties of recycled pulsars**, Universität Bielefeld

Teaching

- 2024** 16-hour PhD course at the University of Milano-Bicocca titled "Stochastic gravitational-wave backgrounds from the nanohertz to the kilohertz."
- 2010–2013** Responsible for guiding and assisting lab interns and students of the Department of Electronic Science, the University of Pune and other colleges on various projects for BE/BTech (3), MSc (4) and MTech (1). This involved approximately 2 hours of hands-on supervision per week for four to six months per student, for a total of about 200 hours of supervision.

Curriculum Vitae

- 2010** Designed and conducted a bridge course for first semester students of MSc (electronics) in collaboration with faculty members at the department. This two day course involved four hours of teaching by me, apart from time spent on the course coordination.
- 2010-2013** Designed and conducted practical laboratory courses for the second, third and fourth semester students of the MSc (Electronics) course at the Department of Electronic Science, University of Pune. Each course was worth 4 credits and required 60 hours of teaching.
- 2010-2011** Taught an Applied Electrodynamics Laboratory course for the fourth year students of the integrated MSc program at the Centre for Excellence in Basic Sciences (CBS), an autonomous institute based on the Kalina campus of the Mumbai University. The course required 36 hours of teaching.

Supervision

- 2025 - present** Co-supervising postdoctoral research work of Dr. Jonas El Gammal (Università degli studi dell'Insubria).
- 2025 - present** Co-supervising postdoctoral research work of Dr. Daniele Spinoso (Università degli studi dell'Insubria).
- 2024 - present** Co-supervising postdoctoral research work of Dr. Eleonora Villa (IASF, INAF-Milano).
- 2024 - present** Co-supervising Master thesis "European PTA data analysis using Discovery" by Federico Fantoccoli (Università degli Studi di Milano-Bicocca).
- 2023 - present** Co-supervising PhD thesis of Irene Ferranti (Università degli Studi di Milano-Bicocca).
- 2023 - present** Co-supervising PhD thesis of Beatrice Moreschi (Università degli Studi di Milano-Bicocca).
- 2023** Co-supervised master thesis on the "Joint analysis of deterministic and stochastic signals on Pulsar Timing Data" by Irene Ferranti (Università degli Studi di Milano-Bicocca).
- 2023** Co-supervised master thesis on the "Searching for eccentric binaries in PTA datasets" by Sara Manzini (Università degli Studi di Milano-Bicocca).
- 2023** Co-supervised master thesis on the "Searching for high frequency continuous gravitational waves with MeerTime" by Beatrice Moreschi (Università degli Studi di Milano-Bicocca).
- 2022** Co-supervised bachelor thesis on the "Constraining equations of state for the cold dense matter of massive pulsars" by Fabrizio Davide Brown (Università degli Studi di Milano-Bicocca).
- 2022** Co-supervised master thesis on the "Establishing a threshold on the Bayes factor for a robust detection of the GWB in PTA datasets" by Serena Valtolina (Università degli Studi di Milano-Bicocca).
- 2021 - 2022** Co-supervised master thesis on the "The effects of outliers on gravitational wave searches in pulsar timing arrays" by Giulia Fumagalli (Università degli Studi di Milano-Bicocca).
- 2019 - 2021** Co-supervised PhD projects on the "Optimizing Analysis Standards for Pulsar Timing Arrays and Gravitational Wave Detection" led by Jun Wang (Bielefeld Universität).
- 2019** Supervised Caroline el Khoury for "Scintillation studies of pulsars in binaries", as part of the ASTRON/JIVE Summer Programme 2019.
- 2018 - 2020** Co-supervising PhD project "Precision timing for pulsar timing arrays" led by Emma van Wateren (Nijmegen University/ASTRON).

Curriculum Vitae

- 2018** Co-supervised PhD project "A study of timing stability of Black Widow pulsars" led by Ann-Sofie Bak Nielsen (Leiden University/ASTRON).
- 2017** Co-supervised bachelor thesis on the "Spectral Indices of Millisecond Pulsars" by Philip Bergjann at the Bielefeld University.
- 2015** Co-supervised bachelor thesis on the "A Statistical Study of Pulsar Proper Motions" by Timo Kaja at the Bielefeld University.
- 2014** Co-supervised bachelor thesis on the "Dispersion measure variations for 80 pulsars on short time scales observed with LOFAR" by Julian Donner at the Bielefeld University.

Tutorials

- Apr, 2022** **Pulsar analysis with Enterprise**, special sessions for the Astronomical Observatory of Cagliari, Cagliari, Italy
- Jun, 2019** **Introduction to PSRCHIVE**, International Pulsar Timing Array student week, Pune, India
- Jun, 2019** **Python bindings for PSRCHIVE**, International Pulsar Timing Array student week, Pune, India
- Jun, 2019** **Pulsar timing with Tempo2**, International Pulsar Timing Array student week, Pune, India
- Nov, 2016** **Regression analysis with python**, Universität Bielefeld

Diversity & inclusiveness

- 2020 - present** Conceived and continue to organise the EPTA *beverage of choice chat* sessions, where collaboration members meet bi-annually to discuss a wide range of topics affecting academic life, from the impacts of online/smart working to neurodivergence.

Grants, honors & awards

- 2016** **MPG PhD Stipendiat - extension**, Max Planck Society, Munich, Germany. Financial support for 6 months worth a total of €8.190,00.
- 2014** **General Radionet funding**, assigned by the Radionet consortium in specific occasions (meetings, conferences). The financial support was of €1.300,00, of which I was the beneficiary.
- 2013** **MPG PhD Stipendiat**, Max Planck Society, Munich, Germany. Financial support for 36 months worth a total of €49.1400,00.
- 2009** **Summer Research Fellowship**, Indian Academy of Science, Bangalore.

Observing grants

Since 2016 I have won hundreds of hours of telescope observing time at the largest radio telescopes in the world. I am a principal investigator or co-PI on successful proposals with the Effelsberg 100-m radio telescope in Germany, the Arecibo radio telescope in Puerto Rico, the pan-European Low Frequency Array (LOFAR) radio telescope, the Giant Metrewave radio telescope in India and the Parkes 64-m radio telescope in Australia. Besides this I am also a member of key science programs at the MeerKAT observatory in South Africa and the NENUFAR observatory in France.

Leadership Roles

- 2024 - present** Member, Gravitational Wave International Committee
- 2022 - 2024** Executive Committee member, the European Pulsar Timing Array
- 2022 - present** General secretary, the Gravitational Wave Early Career Scientists group
- 2022 - present** Co-chair - Data Combination Working Group, the International Pulsar Timing Array
- 2021 - present** Founder chair - PTA Early Career Scientists Working Group, the International Pulsar Timing Array
- 2017 - 2024** Chair - Timing Working Group, the European Pulsar Timing Array
- 2017 - 2024** Chair - IISM Working Group, the European Pulsar Timing Array

Membership

- 2023 - present** LISA Astrophysics Working Group
- 2022 - present** The European Consortium for Astroparticle Theory (EUCAPT)
- 2021 - present** TEONGRAV - Theory of Gravitational Wave Sources (INFN, Italy)
- 2019 - present** NenuFAR Pulsar Key project
- 2019 - present** Polarization Calibration task-force for the LOW Frequency ARray (LOFAR)
- 2019 - present** Solar Physics and Space Weather LOFAR Key Science Project
- 2017 - present** MeerTime project
- 2014 - 2020** Research training group: Models of Gravity
- 2013 - 2017** International Max Planck Radio School (IMPRS)
- 2013 - present** European Pulsar Timing Array (EPTA) collaboration
- 2013 - present** International Pulsar Timing Array (IPTA) collaboration

Workshops organised

- 2025** Early Career Researcher workshop, Glasgow, the UK
- 2025** IPTA 2025 student week: SOC, Training workshop for IPTA students
- 2024** EAS symposium S2 Astrophysics and Cosmology with Pulsar Timing Arrays: SOC
- 2024** IPTA 2024 student week: LOC, Training workshop for IPTA students
- 2024** IPTA 2024 student week: SOC, Training workshop for IPTA students
- 2021** IPTA 2021 student week: LOC, Training workshop for IPTA students
- 2021** IPTA 2021 student week: SOC, Training workshop for IPTA students
- 2021** Workshop on Gravitational-Wave Astrophysics for Early Career Scientists: Lorentz center, planning & chair of the PTA session
- 2019** Coordinator - ASTRON/JIVE summer programme 2019
- 2019** IPTA 2019 student week: SOC, Training workshop for IPTA students
- 2016** Student-speak : Student Organiser, Mini-workshop for talks by PhD students of MPIfR-Bonn & Universität Bielefeld
- 2011** LOC (Lead Organiser), The Solar Radio Workshop at NCRA, Pune, India
- 2010** Student Coordinator, Frontiers in Physics at Fergusson College, State level workshops for teachers and student, Pune, India
- 2009** Student Coordinator, Frontiers in Astronomy at Fergusson College, National level workshops for teachers and student, Pune, India

Meetings organised

- 2025** 16th Edoardo Amaldi Conference on Gravitational Waves (Amaldi16), Glasgow, the UK
- 2024** IPTA Science meeting 2024, Sexten, Italy

Curriculum Vitae

- 2024** EAS Symposium S2: Lighthouses on the Universe: astrophysics and cosmology with Pulsar Timing Arrays, Padova, Italy
- 2023** EPTA fall meeting Oxford, The UK
- 2023** EPTA spring meeting Milan, Italy
- 2020** EPTA meeting Online

Refereeing

The Astrophysical Journal • Physical Review D • The Monthly Notices of the Royal Astronomical Society • Classical and Quantum Gravity • Science China Physics, Mechanics & Astronomy. • Physica Scripta • Proposal reviewer for the Time Allocation Committee of the Giant Metrewave Radio Telescope, India • Swiss National Science Foundation.

Publications

I have published **62** refereed articles, which have been cited **4813** times as of 11 June 2025. My h-index as reported by the SAO/NASA Astrophysics Data System is **27**. Apart from this, I have also published 1 software package, and published 9 other articles, including 1 Nature Astronomy report and 2 conference proceedings. My full publication record is attached separately and can also be found [on my website](#).

Talks

I have given **5** invited talks including plenary talks at the Amaldi15 and the 'Progress on Old and New Themes in cosmology - PONT 2023' conferences. I have also presented **4** seminars and **33** contributed talks. The full list of talks can also be found [on my website](#).

Outreach

- May, 2020** "Pulsars", online lecture for students of the Woodside High School, London, UK.
- Summer, 2019** Coordinator and Guide, ASTRON/JIVE, University of Amsterdam and University of Leiden Summer Programmes coordinated tour of the Westerbork Synthesis Radio Telescope and LOFAR
- Summer, 2018** Volunteer, ASTRON Open Day, SKA activities for children and 'Ask an Astronomer'
- 2013** Student Coordinator, "Dabholkar Memorial Lecture" by Prof. J. V. Narlikar, Pune, India
- 2011-2013** Volunteer, Public stargazing events with small telescopes, Pune, India

Languages Known

English First Language
Hindi Advanced
Bengali Advanced
Marathi Intermediate

German Basic
Dutch Basic
Italian Basic

References

Dr. Alberto Sesana : alberto.sesana@unimib.it

Professor, Dipartimento di Fisica 'G. Occhialini', Università degli Studi di Milano-Bicocca.

Dr. Joris Verbiest : Joris.Verbiest@ucf.edu

Professor, Bielefeld Universität, Bielefeld, Germany.

Dr. Caterina Tiburzi : caterina.tiburzi@inaf.it

Staff Scientist, INAF - Osservatorio Astronomico di Cagliari, Selargius, Italy.